Research Methodology & IPR

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Research Methods $\xleftrightarrow{\text{Techniques \& Technicalities}}$ Research Methodology

A Course for M.Tech. in Biomedical Instrumentation, Computer Science & Engineering, and Optics & Optoelectronics

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1 Overview
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5 B R E A K
6 Data
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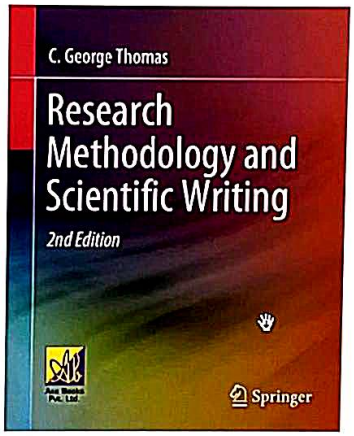
Reference books



Aspects of Research

- Approaches to Research
- Major Research Methods
- Experimental Research
- Collection and Analysis of Data
- Planning and Writing a Research Proposal
- Publications and the Library

Variables in Research

- Variables in Research
  - Independent and Dependent Variables; Important for exploring cause and effect relationships.
  - Qualitative and Quantitative Variables; One can measure quantitative variables but not qualitative variables.
  - Nominal and Ordinal Variables; Example of nominal variable is 'educational status.' Ordinal variables have no absolute values.
  - Continuous and Discrete Variables; Length of a plant is a continuous variable, whereas 'size of a family' is a discrete variable and are integers.
- Formal Approaches to Research
  - Basic Research It has no immediate application. Aim is to enhance knowledge
    - Pure basic research
    - Strategic basic research
  - Applied Research It is for practical applications and benefits.
    - Evaluation research
    - Developmental research

Major Research Methods

- Natural Observation — researchers usually resort to natural observation
- Historical Research — relates past events to one another or to current events
- Ethnographic Research — study of current events rather than past events involving people
- Cross-Sectional Study — in respect of the study population and the time of investigation
- Longitudinal Study — measurements are made over a period, past and present
- Cohort Study — a common trait within a subgroup of population (the cohort) is observed
- Case Study — get a detailed background view of a particular case or phenomenon
- Correlational Research — changes in one or more variables are related to changes in other variables
- Model Building — an equation or formula to estimate a physical process on some other process
- Action Research — Action research encompasses two activities, action and research
- Participatory Learning and Action (PLA) — ordinary people to play an active and influential part
- PLA Tools — various participatory learning and action (PLA) tools

Experimental Research

- Cause and Effect Relationships: Variables in an experiment – the 'cause' and the 'effect.'
- Hypothesis in Experiments: It is a possible or probable answer to a specific question.
- Validity of Experiments: Internal validity (lab), and external validity (real world).
- Principles of Experimentation: Randomization, replication, and local control or blocking
- 'Control' in Experiments: 'control group' and the 'experimental group'
- Classification of Experiments: true expt, quasi-expt, and ex post facto experiments
- Experimental Design: A good design allows valid comparisons between treatments.
- Field Experiments: A true experimental study performed outside the lab
- Randomized Controlled Trial: Experiment involving human subjects with randomized design
- Quasi-experiments: A true expt study without randomly assigned groups
- Ex Post Facto Experiments: An expt without manipulating the independent variables.
- Requirements of a Good Experiment: Well-planned and properly laid out expt

Collection and Analysis of Data

- Sampling: Selecting a sample from a big mass is called sampling
- Scales of Measurement: Nominal scale, ordinal scale, interval scale, and ratio scale.
- Collection of Data: Data collection is an important feature
- Observation: Observe and record information on several features
- Surveys: Survey is a common tool for data collection
- Possible Errors in Research: difference of measurements made and the true value is an error
- Analysis of Data: Analysis of data using different mathematical tools
- Inferential Statistics: differences or associations of data can be analyzed
- Monitoring Research: a system for quality assurance must be followed

Planning and Writing a Research Proposal

- Research Projects: to achieve specific research objectives involving definite tasks
- Finding a Research Problem: A clear idea of the nature of problem to study. It requires, experience, a rigorous literature search via Internet search, daily observations, deliberations in conferences, field studies, etc.
- Analysis of Research Ideas: must think about its feasibility critically
- Planning and Writing a Proposal: to write it down in the form of a project proposal
- Proposal Structure: It must include a title, authors info, abstract and executive summary, introduction, objectives, methodology, conclusion, references, etc.
- Presentation and Evaluation of Proposals: requires patience and time
- Major Funding Agencies: There are national and international funding agencies

Publications and the Library

- Role of Libraries in Information Retrieval: spend time in the library in search of information
- Laws of Library Science: should know the laws and library classifications
- Classification of Publications: Publications in a library are arranged under specific subjects
- Tools for Identifying Literature: tools are available for locating relevant literature
- Types of Publications: magazines, newspapers, online publications, journals, etc.
- Grey Literature: theses, research reports, annual reports, pamphlets, newsletters, etc.
- Microfiche: photographic film containing printed information
- Digital Libraries: online database of texts, still images, audio, video, etc. in digital formats